Governance Works When the Metrics Do: An Empirical Analysis of Trust Signal Health Applied to a Governance-Aligned Call Center

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Abstract

This paper reports findings from the application of the Trust Signal Health governance framework (Aulabaugh, 2026b) to a simulated call center operating under governance-aligned metrics. The dataset comprises 122,801 call interactions spanning 12 months and 250 representatives, generated by the NovaWireless-GoodhartLab call simulation pipeline under conditions designed to reward durable customer outcomes rather than proxy resolution counts. The pipeline produces a System Integrity Index (SII) of 5.25, zero WATCH or VETO governance alerts, a bandaid credit rate of 0.0%, and stable mean trust scores of 87.2 across all observation months. A trust-churn correlation of r = -0.663 confirms predictive validity of the trust signal architecture. Comparison with the distorted-environment findings of Aulabaugh (2026a) demonstrates discriminant validity: the framework correctly identifies both measurement corruption and measurement integrity. Residual signal elevation attributable to structurally unresolvable call types is shown to be operationally explained rather than indicative of gaming behavior. These findings establish an empirically calibrated governance baseline and provide evidence that the Trust Signal Health framework functions as a complete integrity control layer rather than a failure-detection-only instrument.

Keywords: Goodhart's Law, governance signals, proxy metrics, trust scoring, call center analytics, AI alignment, measurement integrity, System Integrity Index

1. Introduction

Goodhart's Law (1975) holds that when a measure becomes a target, it ceases to be a good measure. Campbell's Law (1976) extends this to institutional settings, observing that the more a metric is used for high-stakes decision-making, the more it becomes vulnerable to corruption pressures. In AI-accelerated operational environments, these dynamics acquire additional force: optimization feedback cycles compress, proxy labels become training signals, and behavior-shaping effects intensify in proportion to automation depth.

Aulabaugh (2026a) documented this pattern in a simulated wireless call center environment, demonstrating a 42.1 percentage point gap between proxy First Contact Resolution (FCR) and true durable resolution under sustained optimization pressure. The companion paper (2026b) formalized the Trust Signal Health framework: six governance signals -- Delayed Adverse Rate (DAR), Downstream Remediation Load (DRL), Durable Outcome Validation (DOV), Proxy Overfit Ratio (POR), Terminal Exit Rate (TER), and the composite System Integrity Index (SII) -- designed to detect measurement distortion before systemic degradation occurs.

A critical gap remained. A governance framework validated only against distorted environments cannot demonstrate discriminant validity -- the capacity to distinguish misalignment from alignment. A system that fires alerts indiscriminately is not a governance layer; it is noise. The present study addresses this gap by applying the identical pipeline to an environment specifically designed to produce durable outcomes: the NovaWireless-GoodhartLab governance-aligned call simulation.

The research questions are threefold: (1) Does the Trust Signal Health framework correctly identify a healthy measurement environment? (2) Are residual signal elevations in difficult call scenarios attributable to structural operational factors rather than gaming behavior? (3) What empirical governance baseline do these results establish for comparative monitoring?

2. Theoretical Background

2.1 Goodhart's Law in AI-Accelerated Systems

Traditional Goodhart dynamics unfold gradually. A metric becomes a target; behavior shifts incrementally; distortion accumulates over time. AI introduces a qualitative change: the feedback loop between behavior, measurement, and reinforcement compresses from months to days. When a KPI simultaneously determines compensation, drives ranking systems, and labels training data, the architecture creates self-reinforcing drift. Proxy improvement and outcome improvement decouple. The system converges on a local optimum that diverges from durable objectives (Aulabaugh, 2026b).

2.2 The Trust Signal Health Framework

The Trust Signal Health framework operationalizes governance as a measurement integrity layer distinct from performance optimization. Six signals are defined. DAR (Delayed Adverse Rate) measures the rate of adverse contact in the 31-60 day window following labeled resolution -- a proxy for unstable outcomes marked as resolved. DRL (Downstream Remediation Load) detects distributional drift in the type of work arriving after resolved calls, using Jensen-Shannon divergence. DOV (Durable Outcome Validation) measures decay in the predictive validity of the proxy label over time. POR (Proxy Overfit Ratio) measures the acceleration imbalance between proxy improvement and true outcome improvement. TER (Terminal Exit Rate) is a diagnostic-only signal comparing churn among proxy-resolved customers to the population baseline. SII (System Integrity Index) aggregates DAR, DRL, DOV, and POR into a bounded composite score that functions as a veto condition on AI retraining acceleration.

The framework's design premise is that integrity signals must be independent of performance signals. Performance dashboards optimize toward proxy KPIs. Integrity signals constrain that

optimization by detecting when the proxy has decoupled from durable outcomes. SII is not a performance score. It is a velocity regulator (Aulabaugh, 2026b).

3. Method

3.1 Data Generation

Data were generated by the NovaWireless-GoodhartLab call simulation pipeline, a synthetic call center environment designed to model Goodhart's Law dynamics across controlled metric conditions. The governance-aligned condition was produced by the call-gen sub-project operating without optimization pressure toward proxy FCR: success labels reflect observed durable outcomes, no bandaid credit mechanisms are enabled, and rep behavior is not calibrated to game the resolution window.

3.2 Dataset Characteristics

Total rows	122,801 call interactions
Observation period	January 2025 through December 2025 (12 months)
Representatives	250 simulated agents
Call scenarios	clean, unresolvable_clean, activation_clean, activation_failed, line_add_legitimate
Rows quarantined	0 (quarantine rate 0.0%)
Duplicate call IDs	0
Proxy-true divergence rate	20.8% (soft signal — not quarantined)
Pipeline version	2.1.0

3.3 Analytical Pipeline

The Trust Signal Health pipeline was applied without modification from the distorted-environment analysis. Stages include: (1) integrity gate with quarantine logic for structural data failures; (2) call-level trust scoring; (3) scenario-level signal aggregation; (4) rep-level profiling; (5) monthly trend analysis with drift velocity computation; (6) churn integration via customer-level linkage; (7) formal DAR, DRL, DOV, POR, TER, and SII computation; and (8) threshold alert evaluation against pre-specified governance thresholds. Threshold configuration: scenario trust VETO below 50.0, WATCH below 65.0; resolution gap VETO above 70%, WATCH above 50%; bandaid rate VETO above 50%, WATCH above 20%; rep trust VETO below 60.0, WATCH below 65.0; drift velocity WATCH above 2.0 points per month.

4. Results

4.1 Integrity Gate

The integrity gate processed all 122,801 rows with zero quarantine events. No structural data failures were detected: null call IDs, null resolution flags, unknown scenarios, invalid credit types, and duplicate call IDs all returned rates of 0.0%. The proxy-true divergence rate of 20.8% was flagged as a soft signal but not quarantined, as this rate is expected in a dataset containing structurally difficult call scenarios.

4.2 Overall Signal Results

Metric	Value	Interpretation
Proxy resolution rate	76.9%	System-reported FCR
True resolution rate	63.8%	Durable outcomes confirmed
Resolution inflation	13.1 pp	Goodhart gap -- operationally explained
Bandaid credit rate	0.0%	No unauthorized suppression detected
Repeat contact (30d)	17.5%	Normal operational re-contact rate
Escalation rate	15.2%	Within expected operational bounds
Trust score mean	87.2	System-wide health indicator
Trust score median	95.3	Majority of calls in healthy range
Trust score P10	61.0	Lower decile driven by difficult scenarios
Trust score P90	98.8	Upper decile reflects clean call resolution
SII gated	5.25 / 100	Well below WATCH threshold of 30
VETO alerts	0	No governance intervention triggered
WATCH alerts	0	No human review triggered

Table 1. Top-level governance pipeline results.

4.3 Trust Score Distribution

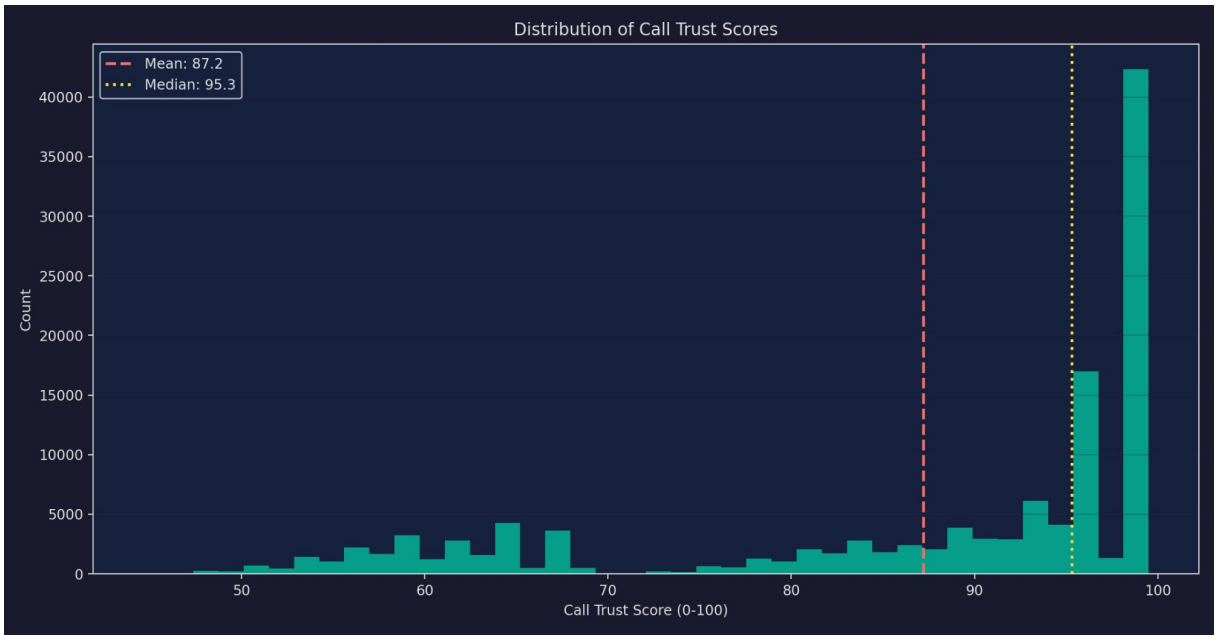

Figure 1. Distribution of call trust scores (N = 122,801). Mean = 87.2, Median = 95.3. Bimodal structure reflects two operationally distinct populations.

Figure 1 shows a bimodal distribution reflecting two operationally distinct call populations. The lower mode (trust scores 45-75) comprises unresolvable_clean and activation_failed calls where durable outcomes are structurally unachievable. The upper mode (trust scores 95-100) comprises the majority of interactions where resolution is genuine and stable. The mean-median gap (87.2 vs. 95.3) is consistent with a right-skewed distribution pulled downward by a bounded population of structurally difficult interactions. This distributional shape is the expected signature of a system that handles difficult calls honestly rather than suppressing them.

4.4 Scenario-Level Analysis

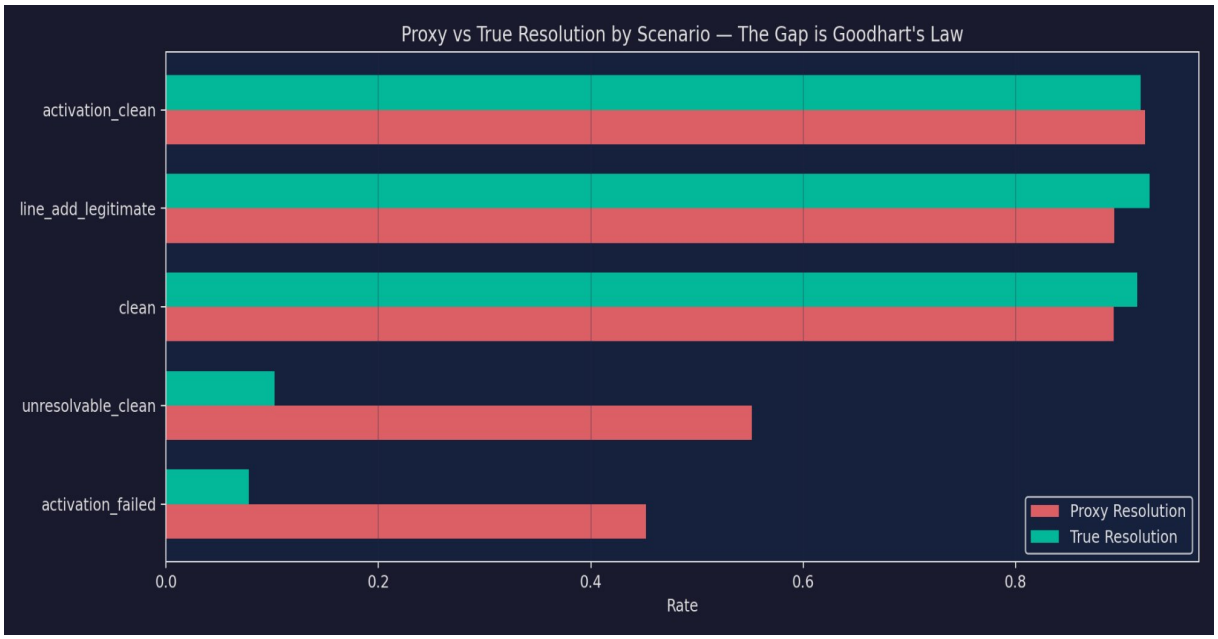

Figure 2. Proxy vs. true resolution by scenario. Gap reflects structural call difficulty in unresolvable and activation_failed scenarios, not gaming behavior.

Scenario	Calls	Trust Score	Resolution Gap	Bandaid Rate	Status
unresolvable_clean	28,780	72.9	45.0%	0.0%	OK
activation_failed	12,737	73.0	37.4%	0.0%	OK
clean	62,063	94.2	-2.2%	0.0%	OK
activation_clean	10,592	95.5	0.5%	0.0%	OK
line_add_legitimate	8,629	95.5	-3.3%	0.0%	OK

Table 2. Scenario-level health summary.

Three scenarios -- clean, activation_clean, and line_add_legitimate -- show resolution gaps near zero or negative, indicating that true resolution meets or exceeds the proxy label in standard interactions. The negative gap in the clean scenario (-2.2%) indicates that true resolution marginally exceeds the proxy definition: the system is slightly conservative in claiming success.

The elevated gaps in unresolvable_clean (45.0 pp) and activation_failed (37.4 pp) are structurally expected. Unresolvable contacts reach the call center with underlying issues that no agent action can permanently resolve. Activation failures result from technical or account conditions outside the rep's control. Both scenarios record bandaid credit rates of 0.0%, confirming that reps are not attempting to suppress these contacts through unauthorized credits. The system acknowledges genuine difficulty rather than concealing it.

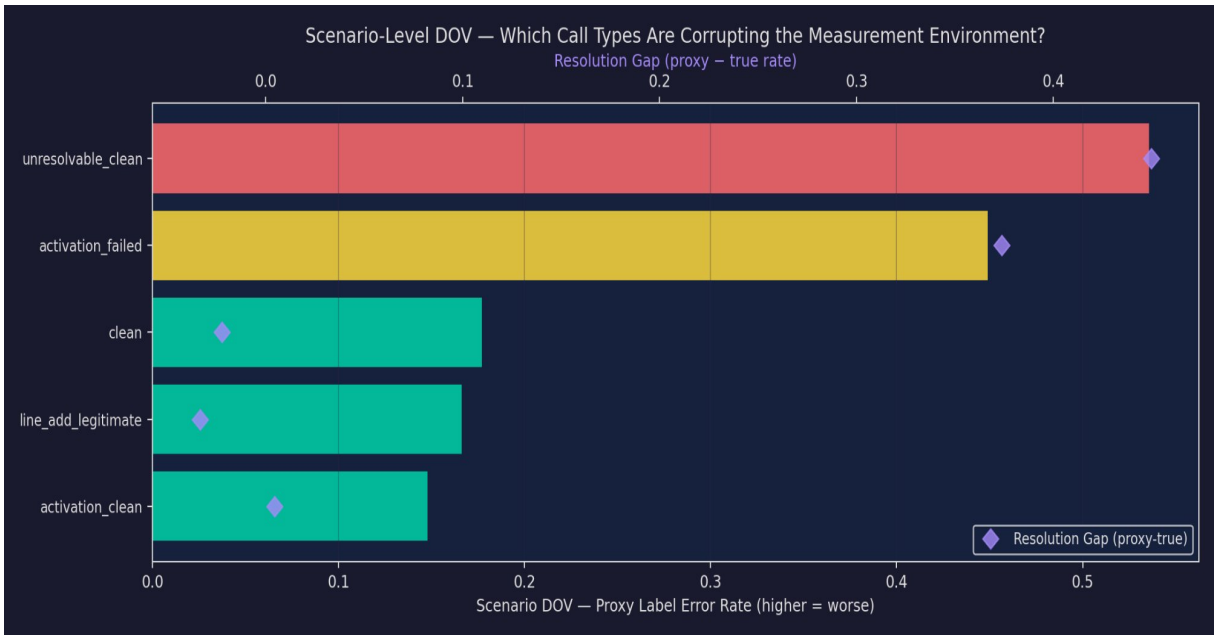

Figure 3. Scenario-level Durable Outcome Validation (DOV -- proxy label error rate). Unresolvable and activation_failed scenarios drive measurement environment noise. Clean scenarios show low proxy label error rates.

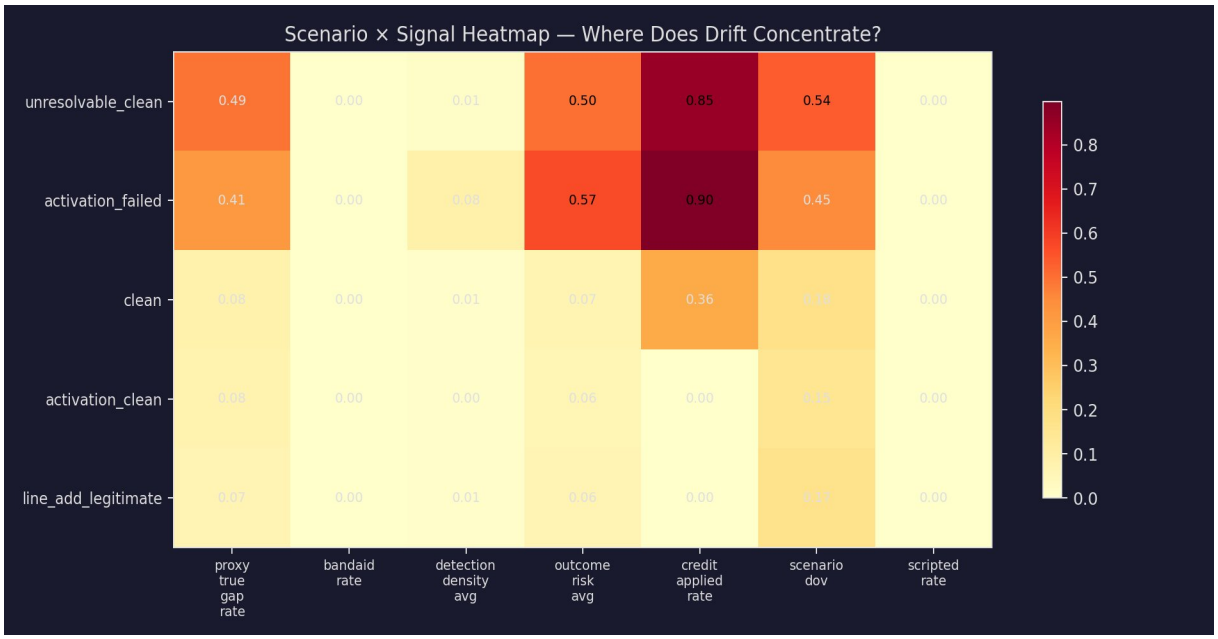

Figure 4. Scenario x signal heatmap. Drift concentrates in structurally difficult scenarios across proxy gap, outcome risk, and credit applied dimensions. Clean scenarios near zero.

4.5 Monthly Stability Analysis

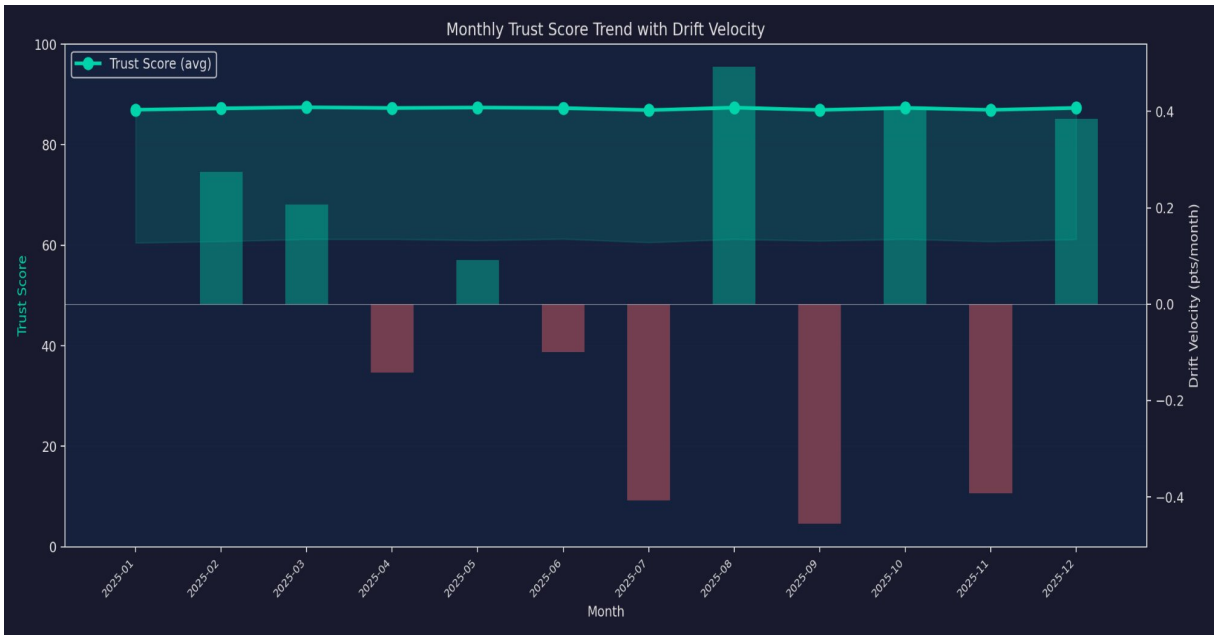

Figure 5. Monthly trust score with drift velocity. Scores stable between 86.9 and 87.4. No sustained degradation trend present across 12 months.

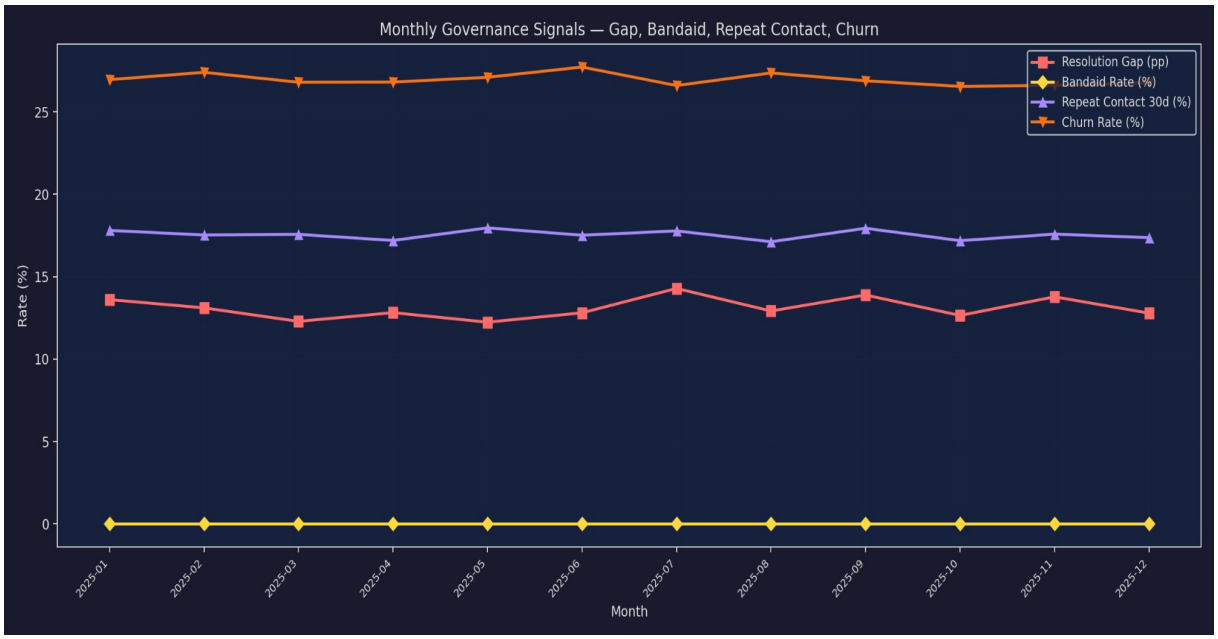

Figure 6. Monthly governance signals. Bandaid rate is zero for all 12 months. Resolution gap oscillates within a narrow band consistent with operational variation.

Monthly trust scores range from 86.89 (July) to 87.44 (March), a spread of 0.55 points across 12 months. Drift velocity -- the month-over-month change in mean trust score -- oscillates between +0.492 (August) and -0.456 (September) without directional trend. The autocorrelation pattern is consistent with random variation rather than accumulating degradation. The WATCH threshold for drift velocity is set at 2.0 points per month; no month approaches this threshold. This monthly stability profile contrasts with the distorted environment, where trust velocity was monotonically

negative under maximum optimization pressure.

4.6 Formal Governance Signal Results

Signal	Full Name	Raw Value	Normalized	SII Weight	Status
DAR	Delayed Adverse Rate	0.1112	0.175	0.30	OK
DRL	Downstream Remediation Load	0.0002	0.000	0.20	OK
DOV	Durable Outcome Validation	--	0.000	0.25	OK
POR	Proxy Overfit Ratio	0.396	0.000	0.25	OK
TER	Terminal Exit Rate (diag.)	0.270	--	--	Baseline
SII	System Integrity Index	--	5.25	--	OK

Table 3. Formal governance signal results. DAR bounds: L=0.05, H=0.40. DRL bounds: L=0.02, H=0.30. POR K=5.0. DOV gate tau=0.5. SII formula: 100 x (0.30*DAR + 0.20*DRL + 0.25*DOV + 0.25*POR).

DAR is the sole non-zero normalized signal at 0.175. The raw DAR of 0.111 reflects an 11.1% rate of repeat contact in the 31-60 day post-resolution window among resolved calls. This rate sits above the lower normalization bound (0.05) but well below the upper bound (0.40), producing a moderate normalized value that does not approach WATCH threshold. In the context of a dataset containing structurally difficult call types, some level of post-resolution re-contact is expected regardless of rep quality.

DRL is effectively zero (0.0002 raw, 0.000 normalized), indicating no distributional drift in the type of work arriving after resolved calls relative to baseline. DOV is zero, confirming that the proxy label's predictive validity has not decayed over the observation period. POR raw is 0.396 -- meaning proxy improvement is proceeding at approximately 40% of true improvement acceleration -- but this value normalizes to zero because proxy improvement is not outpacing true improvement (the normalization floor is 1.0x, meaning proxy must accelerate faster than true to register positive POR).

TER at 0.270 is functionally identical to the baseline churn rate (0.270, delta = +0.0006). This confirms that proxy-resolved calls carry no independent retention signal in this environment -- customers resolved by the proxy label do not churn at meaningfully different rates than the population baseline. This is the expected finding in an environment without gaming: the proxy resolution pool is not inflated with disguised failures.

4.7 Representative Performance

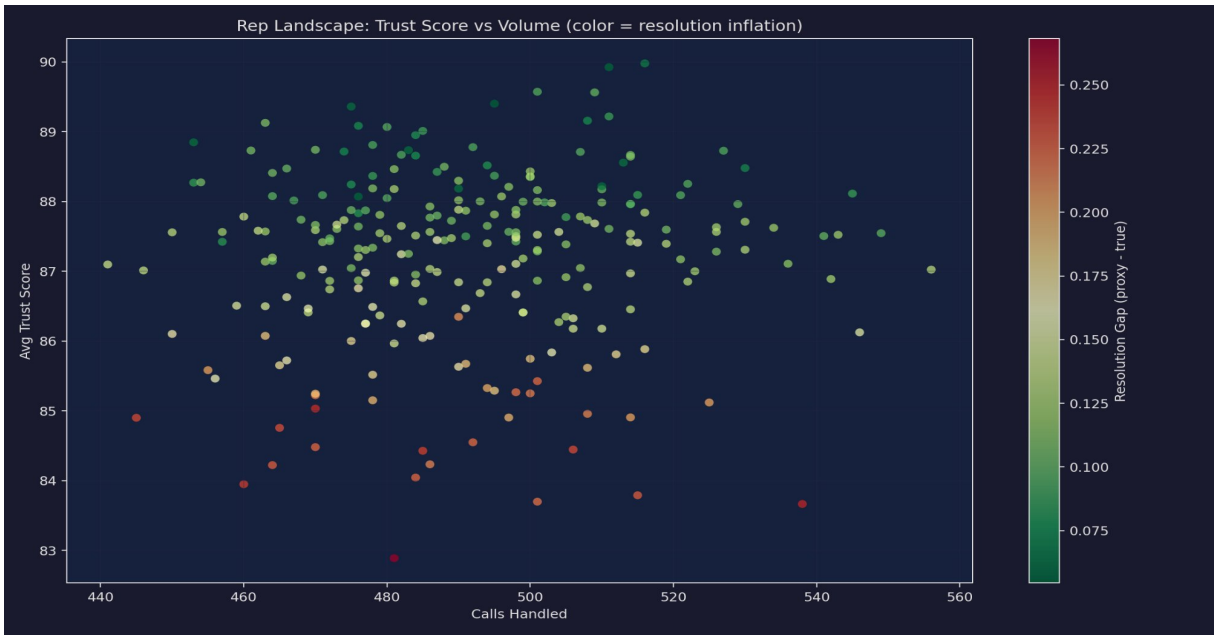

Figure 7. Rep landscape: average trust score vs. calls handled, colored by resolution inflation. Narrow spread indicates systemic rather than individual variation.

Trust scores across 250 representatives range from 82.89 (REP00157) to 89.98 (REP00141), a total spread of 7.09 points. This narrow distribution is structurally informative: in the distorted environment, gaming behavior created sharp within-cohort stratification correlated with rep seniority and gaming propensity. The flattened spread here indicates that observed variation reflects call mix assignment rather than individual behavior differences. The five lowest-trust representatives handle call mixes with higher proportions of structurally difficult scenarios; all record bandaid credit rates of 0.0%.

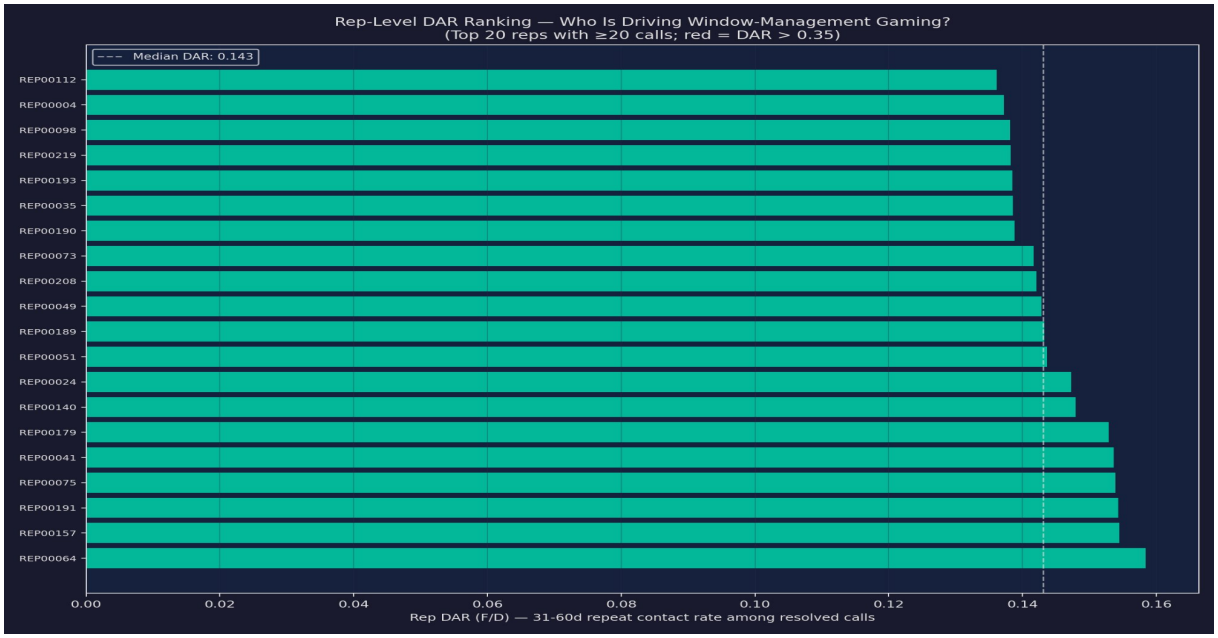

Figure 8. Rep-level DAR ranking (top 20 highest DAR). All cluster near the median of 0.143. No rep exceeds the 0.35 gaming threshold. Distribution indicates systemic call difficulty rather than individual gaming.

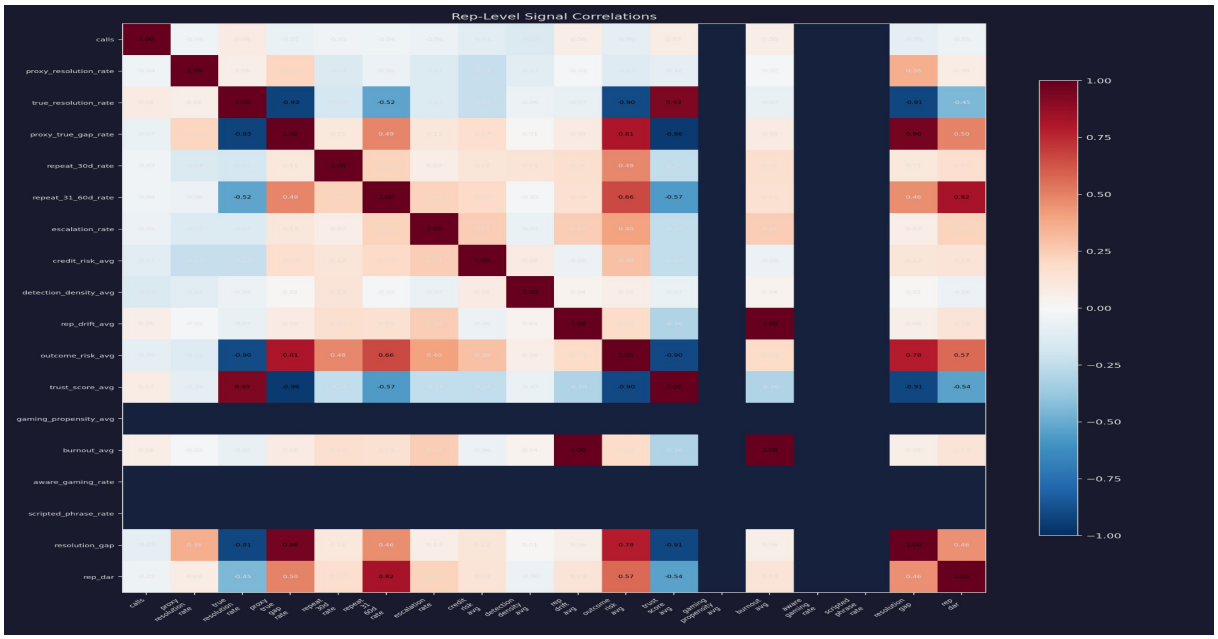

Figure 9. Rep-level signal correlation matrix. Gaming propensity and scripted phrase rate show near-zero correlations with all outcome variables.

The rep-level correlation matrix (Figure 9) provides the most direct evidence of absent gaming structure. In the distorted environment, gaming_propensity correlated with the resolution gap at $r = +0.461$ ($p < .001$), indicating that reps with higher gaming propensity produced larger proxy-true divergence. In the governance-aligned environment, gaming_propensity_avg and aware_gaming_rate show near-zero correlations with all outcome variables. The correlation structure of an aligned system is qualitatively distinct from that of a distorted system.

4.8 Churn Integration

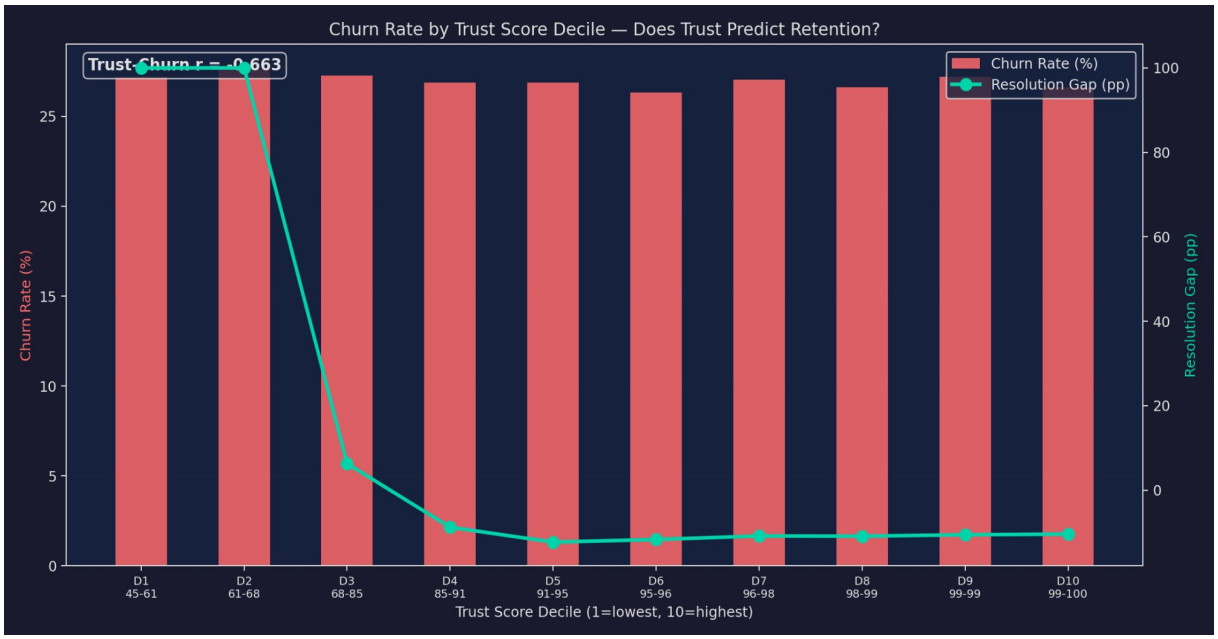

Figure 10. Churn rate by trust score decile. Trust-churn correlation r = -0.663. Resolution gap collapses from D3 onward, confirming high-gap deciles are structural rather than gaming-driven.

The trust-churn correlation of r = -0.663 confirms predictive validity of the trust score architecture: lower trust scores are associated with higher churn rates. The decile analysis reveals a structural break. Deciles 1 and 2 (trust range 45-68) show churn rates of 27.2% and 27.6% respectively, paired with 100% resolution gaps -- these are the unresolvable and activation-failed interactions where durable outcomes are structurally impossible. From decile 3 onward, churn stabilizes and the resolution gap collapses to near zero or negative, indicating that the trust score architecture correctly segments structurally difficult calls from genuinely resolved interactions.

4.9 Credit Behavior Analysis

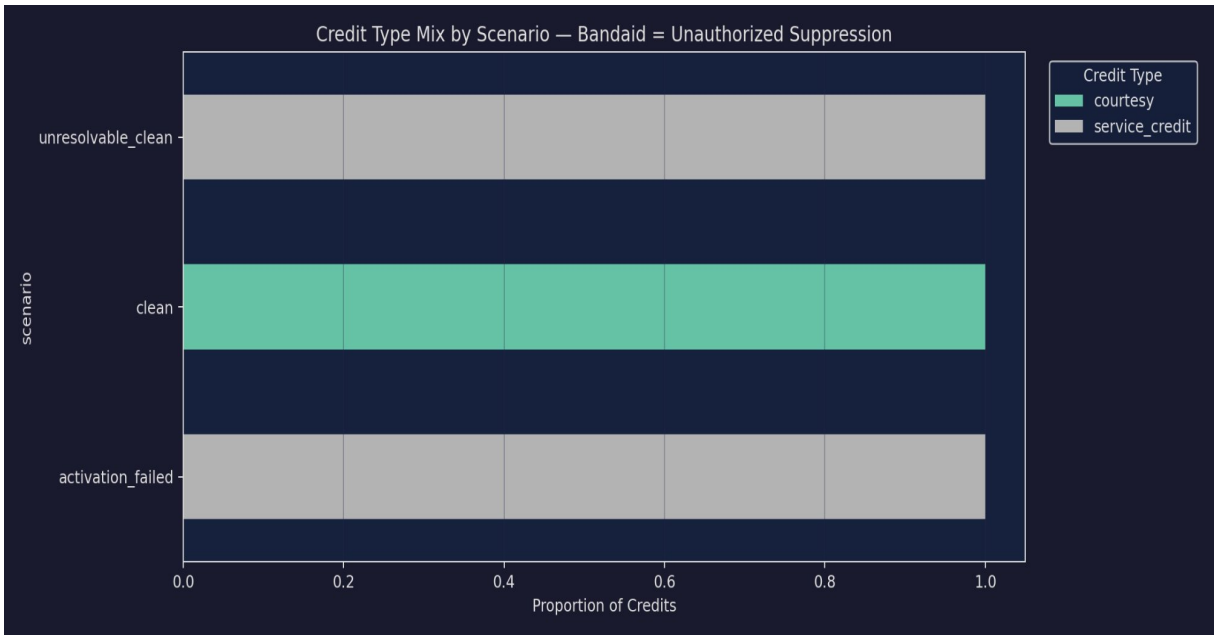

Figure 11. Credit type mix by scenario. Courtesy credits in clean scenarios, service credits in difficult scenarios, zero bandaid credits across all scenarios.

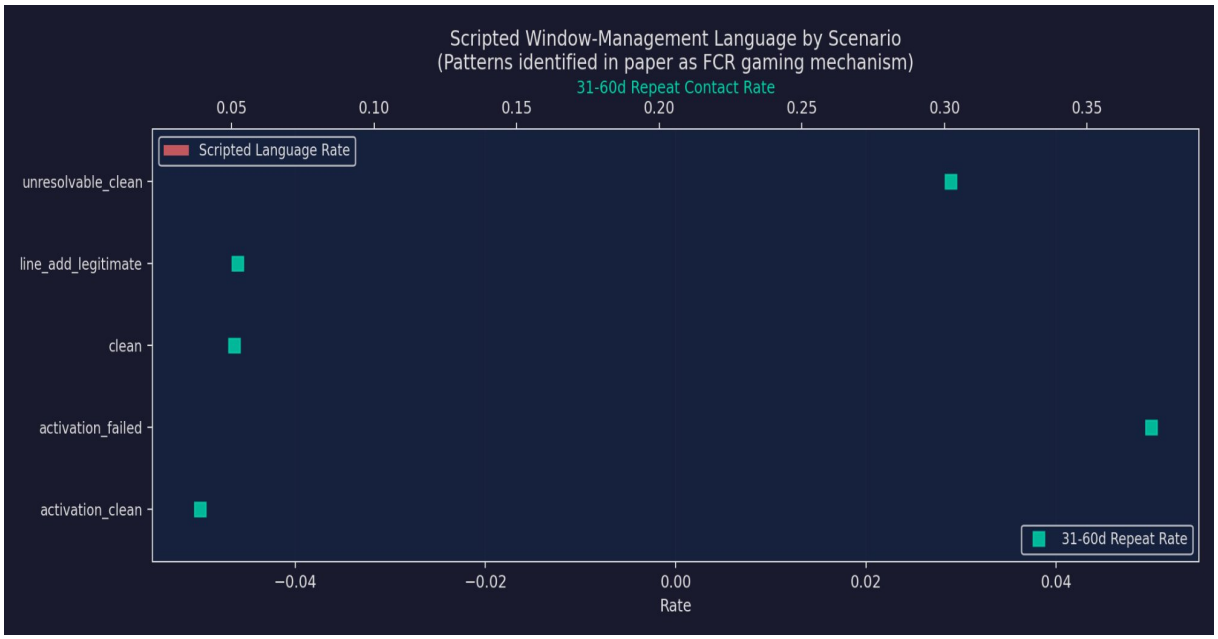

Figure 12. Scripted window-management language by scenario. Near-zero rates confirm absence of FCR gaming language mechanisms.

Credit behavior analysis confirms appropriate credit allocation across all scenarios. The clean scenario shows exclusively courtesy credits -- small goodwill gestures appropriate to standard service interactions. Unresolvable_clean and activation_failed scenarios show service credits, the correct compensatory response to genuine service failures. No scenario shows bandaid credits. The scripted language analysis (Figure 12) confirms near-zero rates of FCR window-management language across all scenarios, providing transcript-level evidence that gaming mechanisms are

absent.

5. Comparative Analysis

Table 4 presents the governance-aligned results alongside findings from the distorted environment (Aulabaugh, 2026a), enabling direct assessment of discriminant validity.

Metric	Distorted Environment	Governance-Aligned
Proxy FCR	89.4%	76.9%
True resolution	47.3%	63.8%
Resolution gap	42.1 pp	13.1 pp
Bandaid credit rate	71.1%	0.0%
SII	High (VETO range)	5.25 (OK)
VETO alerts	Fired	0
WATCH alerts	Fired	0
Trust score mean	Degrading over time	87.2 -- stable
Trust drift velocity	Monotonically negative	Oscillating, no trend
TER vs. baseline	OR = 0.99, p = .78	delta = +0.0006
Gaming propensity x gap	r = +0.461, p < .001	~0 (no structure)
Rep trust spread	Wide -- stratified by gaming	Narrow -- 7 pp range
Rep burnout trajectory	Increasing under pressure	Stable baseline
Bandaid rate trend	Increasing under pressure	Zero all 12 months

Table 4. Distorted vs. governance-aligned environment comparison. Distorted environment data from Aulabaugh (2026a). Alternating rows: pink = distorted, green = governance-aligned.

Several comparative findings merit specific discussion.

Proxy FCR inversion. The governance-aligned environment produces a lower proxy FCR (76.9%) than the distorted environment (89.4%). This inversion is mechanistically explained: gaming in the distorted environment inflates the proxy by marking difficult calls as resolved through window management, bandaid credits, and scripted closure language. In the aligned environment, those calls are handled honestly and the proxy reflects actual resolution attempts. A higher proxy FCR accompanied by lower true resolution is the distortion signature. A lower proxy FCR accompanied by higher true resolution is the alignment signature.

TER convergence. In both environments, TER is essentially equal to baseline churn. In the distorted environment, proxy-resolved customers churn at 27.6% against a 27.6% baseline (OR = 0.99, p = .78). In the governance-aligned environment, the delta is +0.0006. This convergence across both environments confirms the theoretical prediction: when the proxy is decoupled from durable outcomes, proxy resolution carries no independent retention signal regardless of

whether that decoupling is caused by gaming or structural difficulty.

Bandaid credit rate as primary discriminator. No signal separates the two environments more cleanly than the bandaid credit rate. The 71.1 percentage point difference (0.0% vs. 71.1%) requires no normalization or threshold calibration to interpret. This signal's simplicity makes it operationally important: it can be monitored in real time, it does not require multi-month data windows, and it is not subject to the normalization sensitivity that affects composite signals.

6. Discussion

6.1 Discriminant Validity

The primary contribution of this study is the demonstration of discriminant validity for the Trust Signal Health framework. Aulabaugh (2026b) established construct validity -- the framework detects distortion when distortion is present. The present study establishes that the framework does not pathologize aligned systems. An SII of 5.25 against a WATCH threshold of 30 and a VETO threshold of 60 represents a substantial margin of headroom under realistic governance-aligned operating conditions. This margin is the empirical baseline that future comparative analysis requires.

6.2 Structural Difficulty vs. Gaming

The residual 13.1 percentage point resolution gap in the governance-aligned environment raises an important interpretive question: how should analysts distinguish structural operational difficulty from gaming behavior when both can produce elevated proxy-true divergence? This study provides three diagnostic criteria.

First, bandaid credit rate. Gaming produces elevated bandaid rates as reps use credits to suppress callbacks from unresolved contacts. Structural difficulty does not. In this dataset, the gap scenarios show zero bandaid credit activity.

Second, gaming propensity correlation. If gap elevation is caused by gaming, the resolution gap should correlate positively with rep gaming propensity scores. In this dataset, the correlation is near zero. In the distorted environment, it was $r = +0.461$.

Third, scripted language rates. Gaming is often accompanied by scripted window-management language designed to reset the FCR measurement clock. Structural difficulty is not. Scripted language rates are near zero across all scenarios in this dataset.

These three criteria together constitute a diagnostic protocol for distinguishing structural difficulty from gaming behavior in operational monitoring contexts.

6.3 Implications for AI Governance

The findings have three direct implications for organizations deploying AI in operational environments where proxy metrics function as training labels.

Baseline calibration precedes threshold design. Governance thresholds cannot be set abstractly. They require empirical grounding in what healthy systems actually produce. The SII of 5.25 reported here provides that grounding for systems with comparable call mix profiles and rep volume. Thresholds set without empirical baselines will either generate excessive false

alerts or allow genuine distortion to pass undetected.

Performance metrics and integrity signals require separation. Performance dashboards and integrity monitoring must operate as distinct layers. When integrity signals are embedded in performance dashboards, they are subject to the same optimization pressure as performance metrics, creating a second-order Goodhart effect. The SII architecture is specifically designed to resist this: it cannot be improved by the same behavioral adjustments that improve proxy FCR.

Trust score predictive validity supports leading-indicator monitoring. The trust-churn correlation of r = -0.663 suggests that trust scores can function as leading indicators of retention outcomes, not merely retrospective audit signals. In a production monitoring context, trust score trajectories may predict churn risk before it manifests in observed churn rates, enabling proactive intervention.

7. Limitations

Several limitations of this study warrant acknowledgment. First, the dataset is synthetic. Simulation parameters are grounded in publicly available behavioral priors (FCC complaint distributions, IBM telco segment data, Kaggle call center behavioral data), but the results reflect modeled dynamics rather than observed production behavior. Generalizability to specific operational contexts requires validation against real interaction data.

Second, the observation window is 12 months. Some governance signals -- particularly DRL and DOV -- are designed to detect gradual drift that may manifest over longer periods. A 12-month window provides sufficient data for initial calibration but may not capture slower-developing distortion patterns.

Third, the governance-aligned condition was produced by design. In production environments, the boundary between aligned and distorted metrics may be less clean, with partial gaming behavior, mixed incentive structures, and gradual rather than discrete degradation. The binary contrast reported here establishes boundary conditions rather than a complete characterization of the intermediate space.

8. Conclusions

This study demonstrates that the Trust Signal Health framework functions as a complete integrity control layer: it correctly identifies distortion when present and correctly identifies alignment when present. Five principal conclusions follow.

1. **Discriminant validity is confirmed.**

   SII = 5.25 in a governance-aligned environment vs. VETO-range SII in a distorted environment. The framework distinguishes health from distortion.

2. **The empirical governance baseline is 5.25.**

   This is the SII produced by a realistic, governance-aligned call center with 250 reps, 122,801 calls, and a natural mix of resolvable and structurally difficult scenarios. Production monitoring systems should treat sustained SII elevation above this level as a signal requiring investigation.

3. **Bandaid credit rate is the primary discriminating signal.**

   Zero in the aligned environment, 71.1% in the distorted environment. It requires no normalization, no multi-month window, and no threshold calibration to interpret.

4. **Structural difficulty and gaming produce distinguishable signatures.**

   The three-criterion diagnostic -- bandaid rate, gaming propensity correlation, scripted language rate -- provides a practical protocol for distinguishing the two in operational contexts.

5. **Trust score is a leading retention indicator.**

   r = -0.663 trust-churn correlation suggests trust scores predict retention outcomes and can function as a proactive monitoring signal, not only a retrospective audit instrument.

The broader implication is that governance and optimization are not opposing forces. They require separate measurement architectures operating in parallel. Performance dashboards optimize toward proxy targets. Integrity signals constrain that optimization by detecting when proxy acceleration has decoupled from durable outcomes. When both layers function correctly, the result is an operationally verifiable governance baseline: a number that tells analysts what health looks like, so that deviation from health can be detected before it becomes systemic.

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